Techfest IIT Bombay 2026-27 · EV Infra & Green Mobility AI Challenge (MHI × ICAT)

EV Infra Map India

A VAHAN-linked, norm-aware planning engine for public EV charging

Challenge 1: EV Infra Map India · Round 1: Concept, Methodology and Implementation Roadmap

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1. Executive summary

Problem. India's public charging data is fragmented: official counts for the same state differ by up to 2× depending on who is counted. No tool links charger supply to the registered EV fleet or measures compliance with the Ministry of Power (MoP) 2024 siting norms. Planners cannot see where chargers are missing today, or how many are needed by FY 2029-30.

Approach.

- **Demand from VAHAN.** EV registrations (RTO × vehicle class × financial year) are mapped to districts and H3 hexagon cells. Supply is a de-duplicated public charger census built from open data and reconciled with MHI counts.
- **Gap and norm compliance.** A transparent **Charging Gap Index** scores each district on load, access and momentum, and the platform measures compliance with the MoP norms (1×1 km urban grid; 20 km highway spacing).
- **Need by FY 2029-30.** A backtested Holt forecast gives Low, Base and High EV stock, converted into light (<30 kW) and DC (≥30 kW) charge points, with **DC need = max(energy-based, corridor-norm)**.
- **Siting and delivery.** A grid-aware maximal covering model recommends sites. Results are delivered through an open web map and dashboard. Pilot: **Odisha** (30 districts); benchmark: **Karnataka**.

Unique contribution: the first open, reproducible pipeline that links VAHAN EV demand to a de-duplicated charger census, measures MoP 2024 norm compliance, and sizes DC charging by corridor as well as energy.

Headline evidence from our working prototype (Odisha, data read 17 Oct 2026)

1. ~408 EVs per public charging station in Odisha vs ~106 in Karnataka (3.8×). EV stock FY 2025-26 from VAHAN: Odisha 2.47 lakh, Karnataka 7.25 lakh. Station counts from MHI: Odisha 606 (unverified), Karnataka 6,805.

2. Only 24.6% of Bhubaneswar has a public charger within 1 km. There are 19 stations; the MoP 2024 norm implies about 147.

3. Odisha needs about 2,940 public charge points by FY 2029-30 (Base; range 2,510 to 5,380), against about 740 today, roughly 4×. NH-16 has 6 stretches longer than 20 km with no public charger.

Throughout this document, "Done" marks results already produced by our prototype on real data, and "Round 2" marks planned work.

2. Problem and relevance

Charger count (same scope?)	Value	What it actually counts
India, MHI reply, Mar 2026 [7]	27,737 stations	Oil-company (IOCL/BPCL/HPCL) stations only
India, MHI Rajya Sabha reply, Jul 2026 [6]	52,718 stations	All installed stations; no fast/slow split maintained
Karnataka: OMC / BEE 2024 / MHI 2026 [7][9][6]	2,400 / 5,765 / 6,805	Different scope and dates; up to 2.8× apart

- **Counts conflict.** Sources differ on stations vs charge points, public vs captive chargers, installed vs operational, and date. There is no single trusted, geolocated public dataset.
- **EV demand is outpacing chargers.** Odisha new EV registrations rose from 989 (FY 2019-20) to 91,715 (FY 2025-26). The EV share of new registrations is now **9.8%** (Figure 1) [1].
- **Norms are not measured.** MoP Guidelines (17 Sep 2024) [5] set ≥1 public station per 1×1 km urban grid by FY 2030, one every 20 km on both sides of highways, and fast charging every 100 km for heavy vehicles. No public tool reports compliance.
- **Decisions are being made now.** PM E-DRIVE funds 72,300 public stations (₹2,000 crore, to Mar 2028) [10]. Odisha has 170 stations / 688 points approved across 6 cities [11]. Allocation needs district-level evidence.

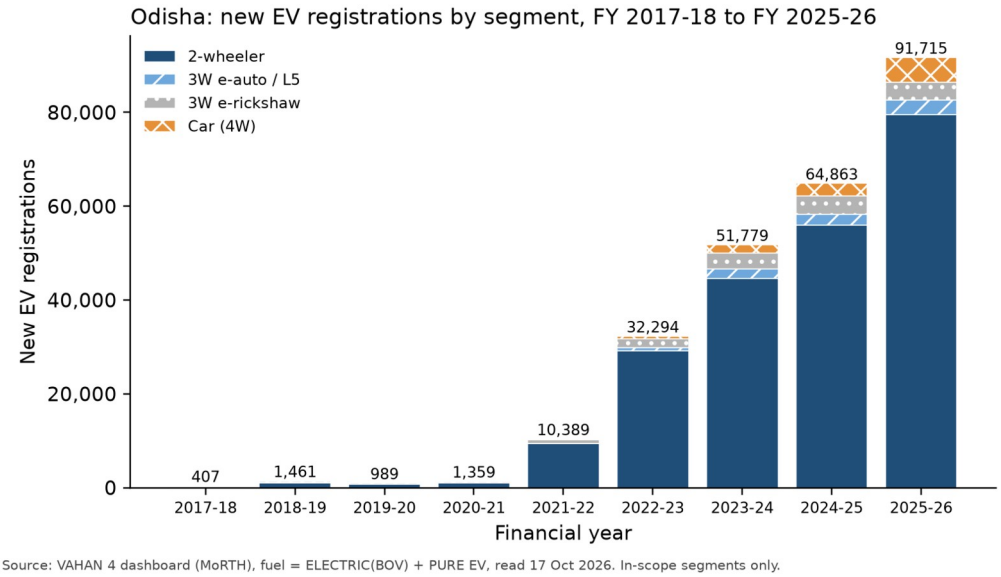


Figure 1. Odisha new EV registrations by segment, FY 2017-18 to FY 2025-26 (in-scope segments).
Source: VAHAN 4 dashboard (MoRTH), fuel = ELECTRIC(BOV) + PURE EV; read 17 Oct 2026 [1].

3. Users and scope

User	Decision	Output they use
MHI / BEE	PM E-DRIVE allocation across states and cities; national tracking	State/district Gap Index ranking; FY30 need (light/DC); norm-compliance %
ICAT	Technical evaluation of planning methods and charger mix	Transparent, reproducible method; validation report
State transport dept.	EV policy targets; PM E-DRIVE site requests	District gaps; ranked candidate sites; 3 scenarios
DISCOM	Grid upgrades, connection capacity	kW added per substation area; grid-aware sites
Urban local body	Public land for charging stations	Uncovered H3 cells inside the city; candidate sites

Scope item	Decision
Geography	Odisha (primary, 30 districts); Karnataka (benchmark, 31 districts); method designed for all states
Units	District (reporting, LGD code) · H3 hexagons (res 8 urban ≈ 0.74 km², res 7 rural) · National Highway segments
Segments	2W; 3W e-auto/L5; 3W e-rickshaw (flagged separately); cars (4W). Buses: depot context layer only
Charger classes	Light (<30 kW: AC + LEV DC) · DC (≥30 kW) · battery swap (separate layer)
Time	Base year FY 2025-26; forecast FY 2029-30 (national 30% EV-sales goal); milestone FY 2027-28 (PM E-DRIVE end)
Out of scope	Home/captive chargers, tariffs, real-time availability, feeder-level power flow, land permits

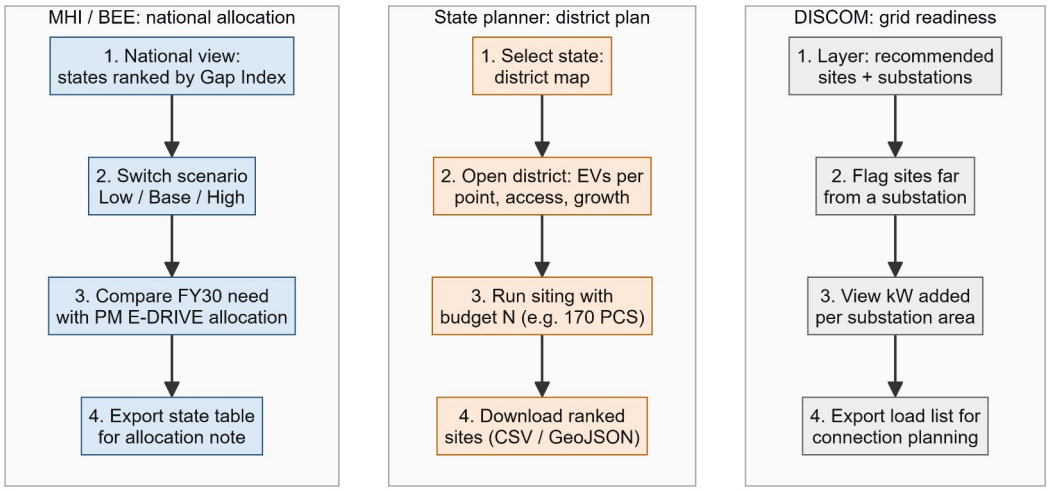


Figure 2. User journeys on the planned dashboard (Round 2).
Source: Team design (Phase 4).

4. Data strategy

Layer	Primary source	Fallback	Granularity	Licence / access
EV registrations	VAHAN 4 dashboard, FY 2017-18 to FY 2025-26 [1]	IndiaDataPortal VAHAN mirror	RTO × class × FY	Govt open data; read in browser
RTO → district	Official VAHAN office list (39 Odisha offices)	State RTO lists	RTO	Built by team
Public chargers	OpenChargeMap export (22 Apr 2026) + OSM [2][3]	Operator locators; EV Yatra spot-check	Point (kW, points)	Provider licences; ODbL
Official counts	MHI / Parliament annexures [6][7]	BEE releases [9]	State	Public
Boundaries	OSM admin_level 5 + LGD codes [3] [26]	Survey of India; DataMeet	District polygon	ODbL
Roads / NH, grid	OSM (Overpass, Geofabrik) [3]	NHAI toll list; OPTCL	Line / point	ODbL
Population, activity	WorldPop 2020 (1 km); GHSL; VIIRS [4][25]	Census 2011 + projections	Raster → H3	CC BY 4.0

Completeness of open charger data (Done)	Our sites	Official stations	Completeness
Odisha (OCM 239 + OSM 2 records; 3 duplicates and 9 closed removed)	229	606 (unverified) [8]	**38%**
Karnataka	415	6,805 [6]	**6%**

- **Better access than expected.** With Y-axis = Rto, X-axis = Vehicle Class and fuel = ELECTRIC(BOV) + PURE EV, VAHAN returns a whole state in **one query per financial year**. Checks pass: segment sums equal VAHAN totals, and Karnataka FY 2024-25 (174,003) matches the 1.73 lakh reported.
- **Synthetic and assumed data** is limited to cell-level EV allocation, swap-station capacity, and planning parameters with no Indian data. Every such value carries an is_synthetic or is_assumed flag (Appendix A2) and is stress-tested in scenarios.

5. Approach overview

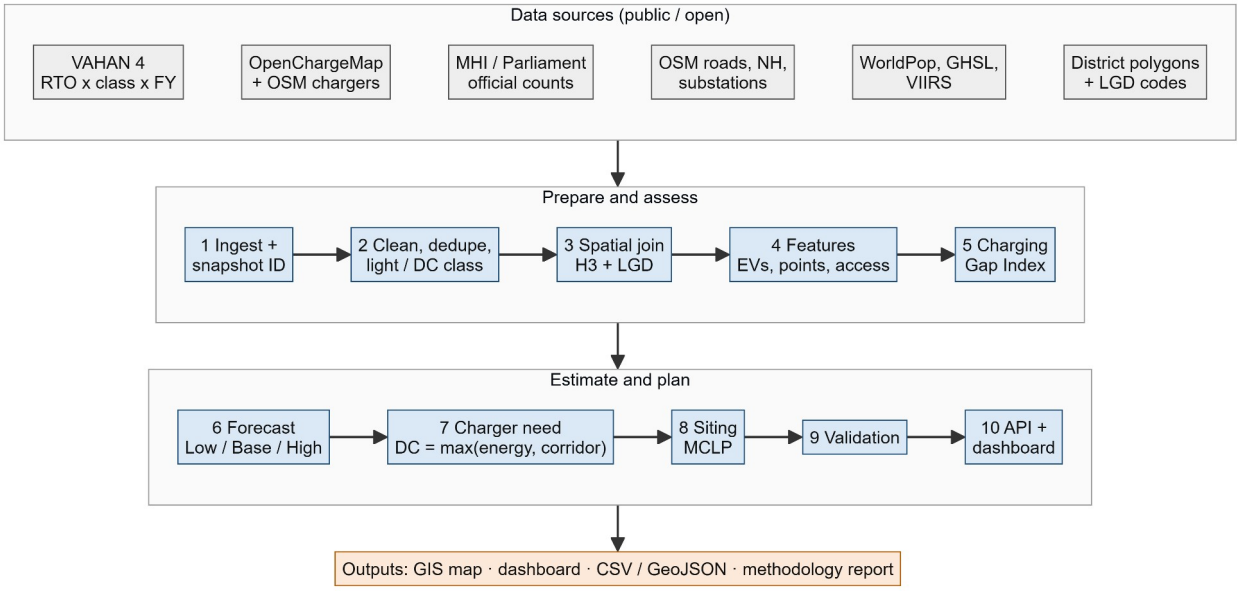


Figure 3. End-to-end pipeline: 10 stages from public sources to planning outputs.
Source: Team design (Phase 4).

#	Stage	Status
1-4	Ingest, clean/dedupe, spatial join, features	**Done** for Odisha + Karnataka (prototype notebook, real data)
5	Charging Gap Index	**Done** (district) · Round 2: H3 cells
6-7	Forecast; charger need	**Done** (state → district, 3 scenarios, backtest) · Round 2: corridor rule for all NHs
8	MCLP siting	Round 2
9	Validation	**Partly done** (backtest, norms, sensitivity) · Round 2: siting hold-out, PM E-DRIVE check

#	Stage	Status
10	API + dashboard	Round 2

6. Data integration

- **Demand mapping:** EV[d,s,t] = Σ_r w[r,d] · REG[r,s,t]. In Odisha all 38 offices with EV records map one-to-one to a district (w = 1); for example, OD-35 Talcher maps to Angul. Offices covering several districts are split by population share and flagged. State-level offices are allocated by each district's share of EVs.
- **Stock with attrition:** STOCK[d,s,T] = Σ_t EV[d,s,t] · (1 − a_s)^(T−t). Annual attrition a: 2W 2%, e-auto 2%, e-rickshaw 10%, 4W 0%.
- **Charger merge:** OCM + OSM (+ operator data in Round 2) go into one schema. **De-duplication** (union-find) merges records that are ≤50 m apart with the same operator, or ≤15 m apart with the same kW class. The most complete record is kept and every source ID is stored.
- **Classification and filters:** light if max kW <30, DC if ≥30. If kW is missing, DC is assumed for a DC connector or a DC-heavy operator (flagged; 36 Odisha records). Private/captive and non-operational records are dropped.
- **Cell-level demand (Round 2):** EV[h] = EV[d] · pop_h · NTL_h^0.5 / Σ(pop · NTL^0.5), using WorldPop population × VIIRS night-lights; flagged as synthetic.
- **Versioning:** every run carries a snapshot ID. Raw files are immutable; processed tables are append-only; the dashboard shows the data date. Refresh: VAHAN monthly, OCM/OSM weekly, MHI per Parliament session.

7. Gap assessment: Charging Gap Index

Component	Definition	Weight	Rationale
Load	ln(1 + EV stock / (observed charge points + 1))	40%	Pressure on today's network
Access	% of population > 5 km from any public charger (1 km for urban cells)	35%	Equity: who cannot reach a charger
Momentum	FY 2023-24 → FY 2025-26 registration CAGR (capped at 95th percentile)	25%	Where demand is accelerating

Gap Index = Σ w_k · 100 · (x_k − min) / (max − min), on a 0-100 scale where higher means more underserved. **Done** for 30 Odisha districts (Figure 4).

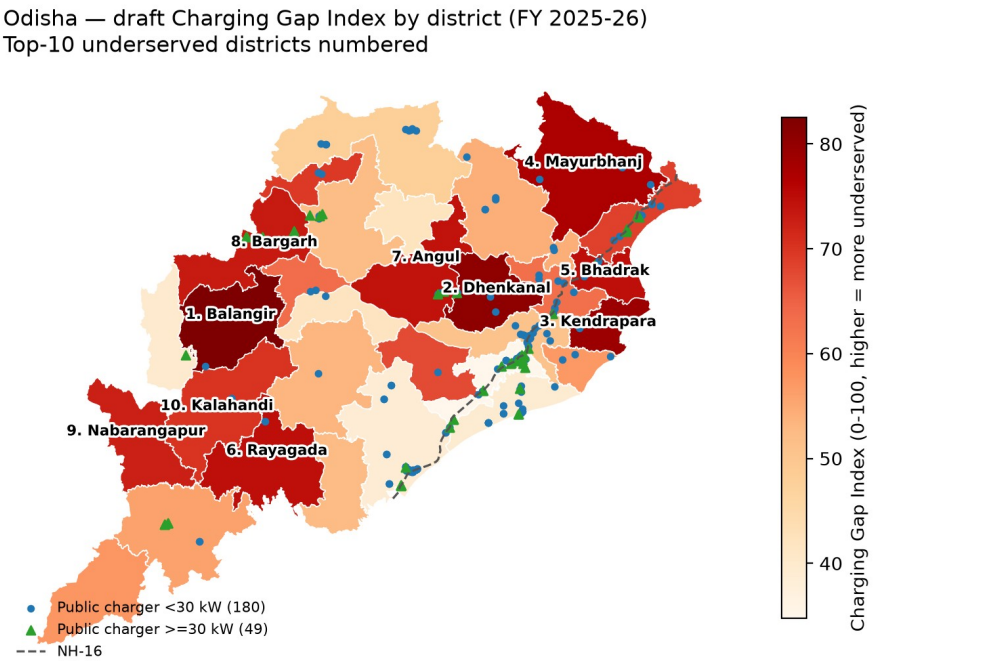


Figure 4. Draft Charging Gap Index by district, Odisha, with public chargers and NH-16.
Source: VAHAN 4 [1]; OpenChargeMap export + OSM, de-duplicated [2][3]; OSM districts; WorldPop 2020 [4]. Team analysis.

Rank	District	Index	EVs FY26	Mapped points	EVs / point	Pop. > 5 km	CAGR FY24-26	FY30 gap (Base)
1	Balangir	82.5	6,004	1	6,004	98%	18%	50
2	Dhenkanal	80.6	3,549	2	1,774	89%	90%	26
3	Kendrapara	79.2	3,547	1	3,547	97%	38%	81
4	Mayurbhanj	77.2	11,417	5	2,283	92%	40%	114
5	Bhadrak	74.7	6,471	6	1,079	85%	90%	64
6	Rayagada	74.5	1,672	2	836	99%	79%	7
7	Angul	74.3	6,849	7	978	87%	86%	76
8	Bargarh	73.1	5,654	4	1,413	95%	43%	50
9	Nabarangapur	72.5	1,322	2	661	97%	90%	13
10	Kalahandi	70.1	2,189	2	1,095	95%	56%	24

Weight sensitivity (Done). We ran 12 weight variants plus a capped-momentum check; results are below.

Variant	Spearman ρ vs Base	Top-10 overlap	Max rank change
Equal weights 33/33/33	0.975	10/10	4
Six ±10-point shifts (min to max)	0.951 to 0.977	8 to 10/10	4 to 6
PCA first-component weights (0.46 / 0.40 / 0.13)	0.968	9/10	6
Leave one out (load / access / momentum)	0.754 / 0.828 / 0.882	8 / 7 / 7	17 / 13 / 8
Momentum capped at 95th percentile	1.000	10/10	0

- **Result.** Every weighting variant passes our rule (ρ ≥ 0.85 and ≥ 8 of 10 districts shared), so the ranking is robust to the choice of weights. Leave-one-out tests move ranks most, which shows each component adds information.
- **Incomplete charger data.** We publish GI_obs and GI_scaled (points ÷ completeness). Maximum rank change is 5, only for Gajapati (no mapped charger). Uniform scaling cannot reveal where the missing chargers are, so Access from observed chargers is an **upper bound on underservice**. Each district carries a data-confidence flag.
- **Note:** the separate ±30% test in Section 10 is on FY30 charge-point gaps, not on this index.

8. Norm compliance (Done)

Norm (MoP, 17 Sep 2024)	Measurement	Result
≥1 public station per 1×1 km urban grid by FY 2030	Bhubaneswar Municipal Corporation (146.9 km²) split into 183 H3 res-8 cells; share of cells with a charger ≤1 km away	**24.6%** of cells covered; 19 stations vs ~147 implied; median distance 1.6 km
A station every 20 km on both sides of highways	NH-16 in Odisha (437 km OSM centreline); gaps between chargers within 2 km of the road	**6 gaps > 20 km** (max 49.4 km); 96 chargers near the corridor
Fast charging for long-range / heavy vehicles	Same corridor, DC chargers only	**7 gaps > 20 km** (max 78.4 km); 24 DC chargers

NH-16 length is from OSM and is shorter than the official length (≈480 km+), so the gap counts are lower bounds.

9. Demand forecast

- **Model:** Holt damped additive trend per segment on Odisha annual registrations, FY 2019-20 to FY 2025-26: $\hat{y}(t+h) = \ell_t + (\varphi + \ldots + \varphi^h) \cdot b_t$. Damping reflects an adoption curve that is levelling off. Parameters are interpretable; seven annual points do not justify ML models.

Backtest: train FY20-FY24, test FY25-FY26 (Done)	Holt damped MAPE	Log-linear MAPE	Decision
2W / 4W	6.2% / 16.9%	488% / 1,276%	Holt
3W e-auto / e-rickshaw	48.4% / 45.8%	334% / 172%	Holt, ±50% band
All (weighted by volume)	**10.2%**	**507%**	**Holt adopted**

Scenario	Definition	EV stock FY 2029-30 (2W/3W/4W)
Low	FY24→FY26 CAGR g_0; growth in year t = $g_0 \cdot 0.5^t$	6.8 lakh
Base	Holt damped trend	**8.3 lakh**

Scenario	Definition	EV stock FY 2029-30 (2W/3W/4W)
High	EV share of new registrations rises linearly to 50% by FY30 (draft Odisha EV Policy 2025, not yet notified); segment markets held at FY26 level	14.4 lakh

- FY 2025-26 stock is 2.47 lakh. The weak 3W forecast matters little because 3W is under 10% of public energy (Base). District allocation now uses each district's FY26 share; Round 2 adds a hierarchical district model reconciled to the state total.

10. Charger requirement

```
E_pub[d] = Σ_s  STOCK[d,s] × km_s × (Wh/km)_s / 1000 × share_s          (kWh/day)
N_light  = E_light / (7.4 kW × 6 h)        N_DC,energy = E_DC / (60 kW × 4 h)
N_DC,corridor = Σ_NH ⌈ L_NH / 20 km ⌉ × 2 sides      (+ ⌈L/100⌉ × 2 heavy-duty)
N_DC = max(N_DC,energy , N_DC,corridor)     Gap = max(0, N − existing_scaled)
```

Parameter	2W	e-auto	e-rickshaw	4W	Source
km/day	25	100	70	35*	Ather, CSTEP, WRI [20][16]
Wh/km	33	60	77	123	LBNL, WRI [15][16]
Public-charging share	10%	30%*	2.5%*	20%*	Kazam-AEEE, IRADe [17][18]
Share at DC	0	0	0	60%*	Assumed

** = assumed (no Indian data); stress-tested ±30%. Charger: light 7.4 kW × 6 h/day, DC 60 kW × 4 h/day (planning targets, assumed).*

Worked example: Baleshwar, Base FY 2029-30	Value
EV stock FY26 → FY30	2W 14,225 → 48,493 · e-auto 1,315 → 3,892 · e-rickshaw 2,770 → 4,481 · 4W 397 → 2,019
Public energy	4,001 + 7,005 + 604 + 1,738 = **13,348 kWh/day**
Light points needed	12,305 ÷ (7.4 × 6) = **277**
DC: energy vs corridor	energy 1,043 ÷ (60 × 4) = 4.3 · corridor: NH-16 ≈116 km in district → ⌈116/20⌉ × 2 = 12
DC need = max(4.3, 12)	**12 DC sites**
Existing (scaled to official count)	light ≈ 32 · DC ≈ 11 (12 and 4 mapped ÷ 0.38)
Gap	**Light ≈ 245**. DC ≈ 1 by count, but the **location gap** remains: NH-16 has stretches of up to 49 km with no charger

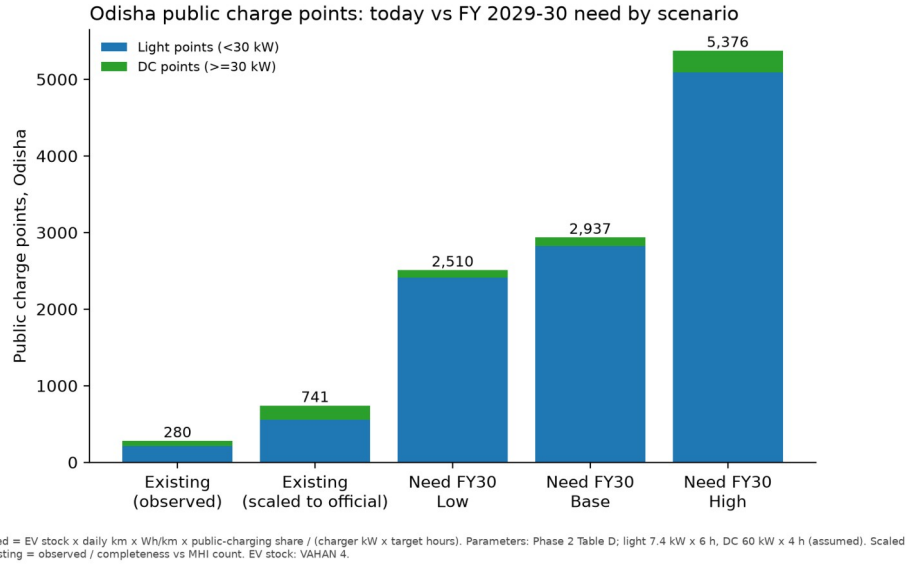


Figure 5. Odisha public charge points: today (observed, and scaled to the official count) vs FY 2029-30 need by scenario.
Source: Team analysis; parameters as in the table above; EV stock from VAHAN 4 [1].

- **State level (Base):** 2,822 light + 115 DC (energy-based) are needed, against about 553 + 188 existing (scaled). NH-16 alone needs ⌈437/20⌉ × 2 = **44 DC sites**, so in Odisha **the corridor rule drives DC need**.

- **Sensitivity (Done):** ±30% on public-charging share or utilisation keeps **9-10 of the top-10 gap districts** unchanged (total gap 1,468 to 3,536 points).

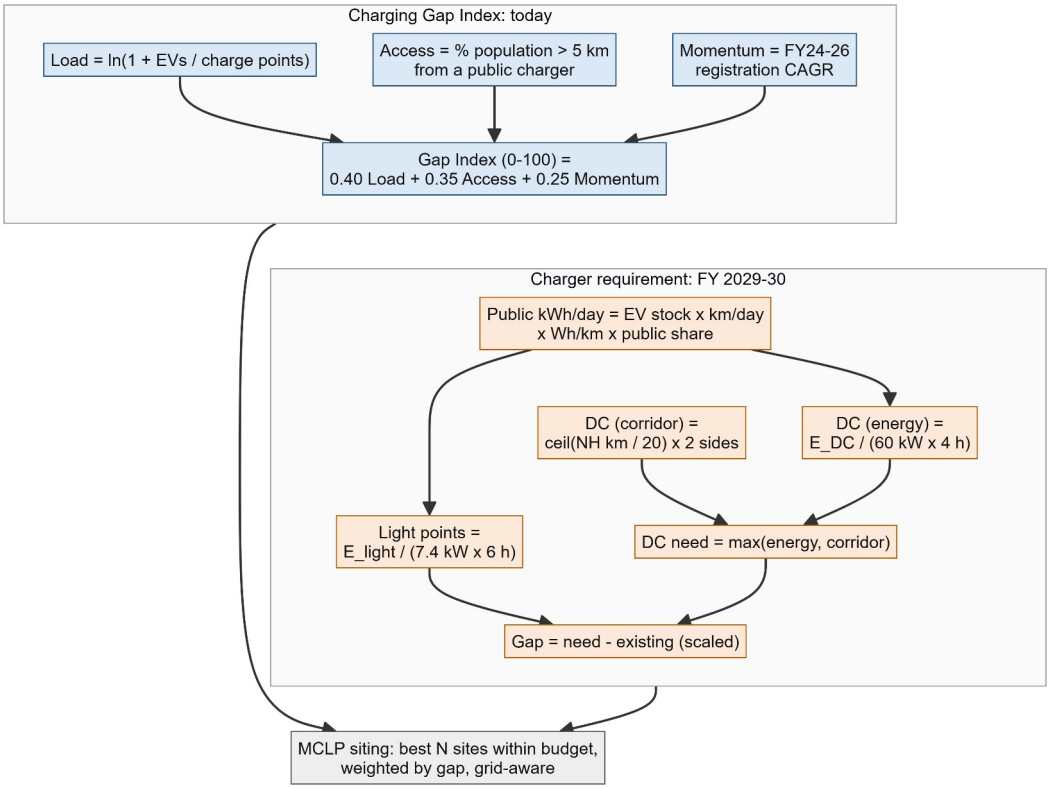


Figure 6. Logic of the Gap Index and the charger requirement, feeding siting.
Source: Team design (Phase 4).

11. Siting (Round 2): Maximal Covering Location Problem

Sets: I = demand cells (H3); J = candidate cells (road access + fuel station, mall, car park, government office, transit hub, PM E-DRIVE site).

Coverage: $N_i = \{ j : dist(i, j) \le R \}$, with R = 1 km urban, 5 km rural, 20 km corridor.

Variables: $x_j \in \{0,1\}$ (open a station at j); $y_i \in \{0,1\}$ (cell i covered).

$$\text{maximise} \quad \Sigma_i \, w_i \cdot y_i \; - \; \lambda \cdot \Sigma_j \, g_j \cdot x_j$$

$$\text{s.t.} \quad y_i \le \Sigma_{j \in N_i} x_j \qquad \Sigma_j \, x_j \le N \qquad x_j = 1 \text{ for existing sites}$$

Inputs: w_i = unmet public kWh/day (Section 10). g_j = distance to the nearest substation (grid penalty), or a hard limit of 5 km. N = budget, e.g. Odisha's 170 PM E-DRIVE stations. Corridor sites use a 1-D greedy with a maximum spacing of 20 km.

Solver: PuLP + CBC with a 5-minute limit (~10^4 cells). **Greedy fallback** adds the site with the largest marginal coverage; it is guaranteed to reach at least 63% of the optimum.

- **Why MCLP:** it answers the planner's question ("best N sites for this budget") and maps directly to the MoP coverage norms. **p-median** needs origin-destination trip data, which is not available. **TOPSIS** has no budget or coverage logic and uses subjective weights, so it is kept only as a tie-breaker.

12. Validation

Test	Design	Status / pass rule
(a) Forecast backtest	Rolling origin: FY20-24 → FY25-26; FY20-23 → FY24-25	**Done:** MAPE 10.2% (pass < 20%)
(b) Norm compliance	Urban 1 km on H3 res 8; NH gaps > 20 km	**Done:** Bhubaneswar 24.6%; NH-16: 6 gaps; reported as measurement
(c) Siting hold-out	Hide 30% of existing chargers, rerun MCLP, measure share recovered within 1/5 km (20 runs)	Round 2. A **demand-model sanity check**, not accuracy: existing chargers are city-biased, while our index targets equity. Pass: ≥ 2× random placement
(d) Index sanity	Overlap of top-quartile districts/cells with PM E-DRIVE (Odisha 170 approved in 6 cities) and state SOP sites	Round 2. Low rural overlap is expected and reported, not tuned away
(e) Sensitivity	One-at-a-time ±30% on each assumed parameter; Low/Base/High; weight variants	**Done** for share, utilisation and weights; Round 2: tornado chart

13. Architecture and tech stack

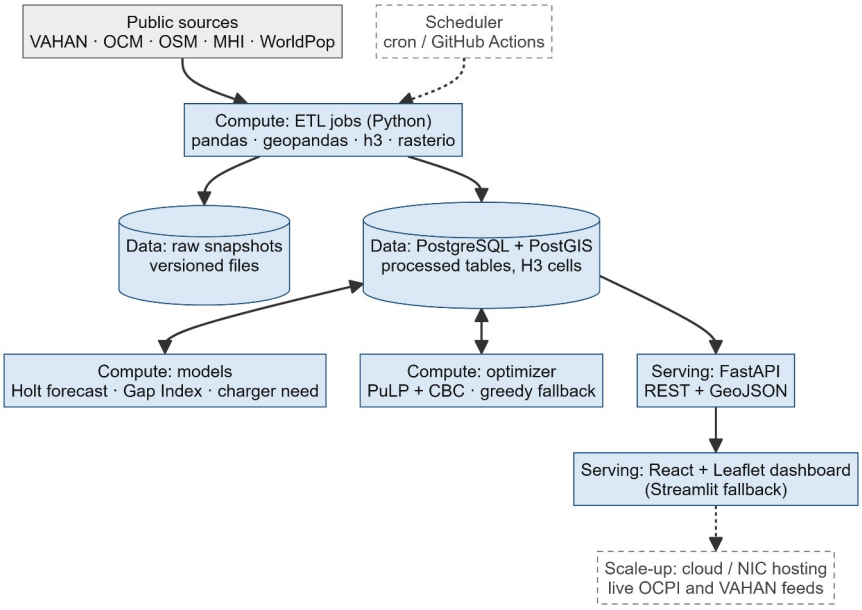


Figure 7. System architecture. Solid = built in Round 2 on a laptop; dashed = future scale-up.
Source: Team design (Phase 4).

Layer	Tool (free / open source)	Why / alternative
Data	PostgreSQL + PostGIS (+ h3-pg); versioned raw files	Spatial SQL that scales nationally / DuckDB-spatial
Processing	Python: pandas, geopandas, h3, rasterio	Same code as the working prototype / polars
Models	statsmodels (Holt); PuLP + CBC	Interpretable, backtested; free MILP solver / OR-Tools
API	FastAPI (REST + GeoJSON)	Typed, auto-documented / Flask
Frontend	React + Leaflet, Recharts; OSM / MapLibre basemap	Light, works offline / Streamlit fallback
Ops	Git/GitHub, Docker Compose, cron / GitHub Actions	One-command demo; reproducible refresh

14. Implementation roadmap (7 Nov to 16 Dec 2026)

Week	Dates	Tasks	Output	Deliv.	Risk
W1	7-13 Nov	Data hardening: MHI annexure, operator locators, EV Yatra spot-check, VAHAN reconciliation; PostGIS schema; refresh scripts	Verified charger census; versioned database	D3 base	Operator data blocked
W2	14-20 Nov	Gap Index final (H3 + district); hierarchical forecast; charger-need module with DC = max rule for all NHs	Need tables by scenario	D4	3W forecast error
W3	21-27 Nov	MCLP siting + substation layer; validation suite (hold-out, PM E-DRIVE overlap, tornado)	Ranked sites; validation report	D2, D4	Solver time
W4	28 Nov-4 Dec	FastAPI endpoints; React + Leaflet map; dashboard with scenario switches and downloads	Working web platform	D1, D2, D5	Front-end slip
W5	5-11 Dec	Karnataka run (scalability test); final methodology document; scalability roadmap	Two-state platform; documents	D3, D7	Karnataka data gaps
W6	12-15 Dec	Demo video; buffer; rehearsal; finale 16-18 Dec	Video; final deck	D6	None (buffer)

Week	Go / no-go milestone	Fallback if missed
W1	Charger census ≥ 60% of official Odisha count; VAHAN gap explained	Keep completeness scaling + confidence flags
W2	Need module reproduces Phase 3 totals; H3 index computed	District-only index
W3	MCLP solves Odisha in < 5 min	Greedy siting (≥ 63% of optimum)
W4	Map + 3 dashboard views live on laptop	Streamlit dashboard
W5	Karnataka pipeline runs without code changes	State-level Karnataka benchmark only

Week	Go / no-go milestone	Fallback if missed
W6	Demo video recorded	Live demo script only

Workstream	W1	W2	W3	W4	W5	W6
Data hardening and database						
Gap Index, forecast, need (D4)						
Siting and validation						
Web platform, map, dashboard (D1, D2, D5)						
Karnataka run, documents (D3, D7)						
Demo video (D6), rehearsal						

15. Assumptions, risks and mitigations

Item	Value used	Stress test	Round 2 fix
Charger completeness	Odisha 38%, Karnataka 6%	GI_obs vs GI_scaled (max rank change 5); missing chargers may cluster in cities, so Access is an upper bound	Operator locators; EV Yatra; MHI annexure
Odisha official count	606 stations (unverified)	Headline 1 shown with ±20% band	Download MHI / Parliament annexure
VAHAN stock vs state figure	1.64 lakh registrations to FY25 vs 2.19 lakh reported (≈25% lower)	High-stock variant (×1.25)	Reconcile with Odisha Transport Dept.
4W public-charging share	20% (range 10-35%)	±30%: top-10 gap districts 9-10/10 stable	Operator session data
4W share at DC	60%	40-80%	Charge-point operator data
Utilisation hours	Light 6 h, DC 4 h	±30%; 2-8 h range	Calibrate to observed 1-5% utilisation
Access threshold	5 km rural, 1 km urban	3 / 5 / 10 km	Travel-time isochrones
High scenario	50% EV share by FY30 (draft policy)	Reported as an upper bound, not a forecast	Update when policy is notified

16. Innovation and scalability

Novel element	Gap it fills in existing platforms
VAHAN-linked demand by segment (RTO → district → H3)	EV Yatra, OCM and operator apps show supply only; none links chargers to the registered EV fleet
Norm-compliance measurement (1 km urban; 20 km corridor)	MoP 2024 norms exist but no tool measures them; we report Bhubaneswar 24.6% and 6 NH-16 gaps
DC need = max(energy, corridor)	Energy-only estimates understate highway DC where cars are few (Odisha: 115 energy-based vs 44 sites on NH-16 alone)
Grid-aware MCLP with a PM E-DRIVE budget	Academic siting is single-city and ignores grid capacity; ours is budgeted, grid-aware and reproducible
Transparent scenarios and completeness flags	Official counts differ by up to 2×; every figure here has a source, snapshot date and assumption flag

Stage	Scope	Enabler
Pilot (Round 2)	Odisha: district + H3, all modules	Working prototype; open data
Benchmark	Karnataka with the same code	One VAHAN query per state per year
National	All 36 states/UTs	PostGIS + scheduled refresh; LGD-keyed districts
Live	Monthly VAHAN, live operator feeds, PM E-DRIVE tracking	OCPI / EV Yatra integration; NIC hosting

A1. Appendix: Sources

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Web sources accessed 17 Oct 2026. Data licences: VAHAN (Government open data); OSM (ODbL); OpenChargeMap (per data provider); WorldPop, GHSL, VIIRS (CC BY 4.0).

A2. Appendix: Synthetic and assumed data register

Item	How it is modelled	Label in data
RTO split for multi-district offices	Population share of jurisdiction (none needed in Odisha; Chandikhole mapping to be confirmed)	is_assumed
Cell-level EV demand (H3)	District EVs × population × night-lights^0.5	is_synthetic
Charger kW class when missing	DC connector / operator rule (36 Odisha records)	is_kw_imputed
Charge points per site when missing	Median by class (1 point)	is_points_imputed
Existing chargers scaled to official count	Observed ÷ completeness (0.38 Odisha)	is_assumed
Planning parameters without Indian data	4W km/day; 3W/4W public share; DC share; utilisation; charger kW; attrition	is_assumed (parameter table)
District allocation of FY30 forecast	FY26 district share per segment	is_assumed
Battery-swap capacity	90 swaps/day × 2 kWh (press-release estimates) [23]	is_synthetic

A3. Appendix: AI-tool disclosure

- **Claude (Anthropic)** was used for research synthesis (finding and summarising public sources), code assistance (data-processing notebook and diagram rendering) and drafting this document.
- All numbers were computed by our code from the cited data. The notebook, raw data snapshots and data-quality log are kept for audit. All figures and decisions were reviewed by the team.
- No proprietary or confidential data was used. Third-party open-source libraries: pandas, geopandas, h3, rasterio, statsmodels, scipy, matplotlib, Mermaid, docx.